



**TRƯỜNG ĐẠI HỌC
CÔNG NGHỆ ĐÔNG Á**
EAST ASIA UNIVERSITY OF TECHNOLOGY

International Training & Cooperation Institute

Introduction to Blockchain Technology

Lesson 1 - Concepts & Terminology



Concepts & Terminology



- 1. What is a Blockchain**
- 2. Key characteristics**
- 3. Cryptocurrencies**
- 4. Consensus Mechanism**
- 5. Mining**
- 6. Crypto Wallet**
- 7. Smart Contracts**
- 8. Discussion**

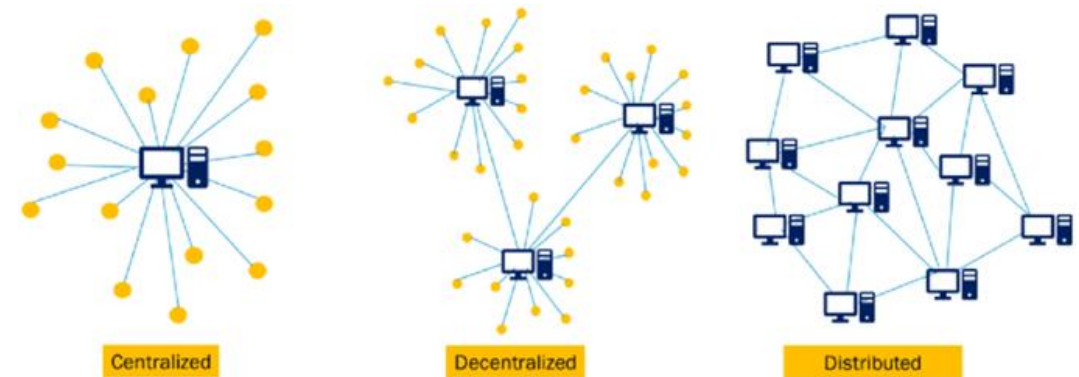
1. What is a Blockchain?

- ❖ A blockchain is a distributed and immutable ledger that is shared among the nodes of a computer network.
- ❖ The innovation with a blockchain is that it guarantees the fidelity and security of a record of data and generates trust without the need for a trusted third party.

=> Blockchain was introduced to solve **trust problems** in digital systems.

❖ How?

- Makes transactions **transparent** and **tamper-evident**
- Solves the **double-spend** problem
- New currency units are created only according to **predefined** rules



Q: Why is trust a problem in digital systems?

1. What is a Blockchain?

- ❖ Software Protocol: Like SMTP is for email, blockchain is a software protocol that defines rules for node communication and data validation
- ❖ The **Existence-Engine**: Its primary role is to provide a distributed **proof-of-existence** certifying that a record existed at a specific time
- ❖ Tamper-Evidence: Because blocks are cryptographically connected, any change to a past record is immediately obvious to all nodes

Q: Is blockchain a cloud-based database?

A: No. It is an application installed on client computers (nodes) that must be connected to the internet to synchronize the ledger

1. What is a Blockchain?

Blockchain Structure

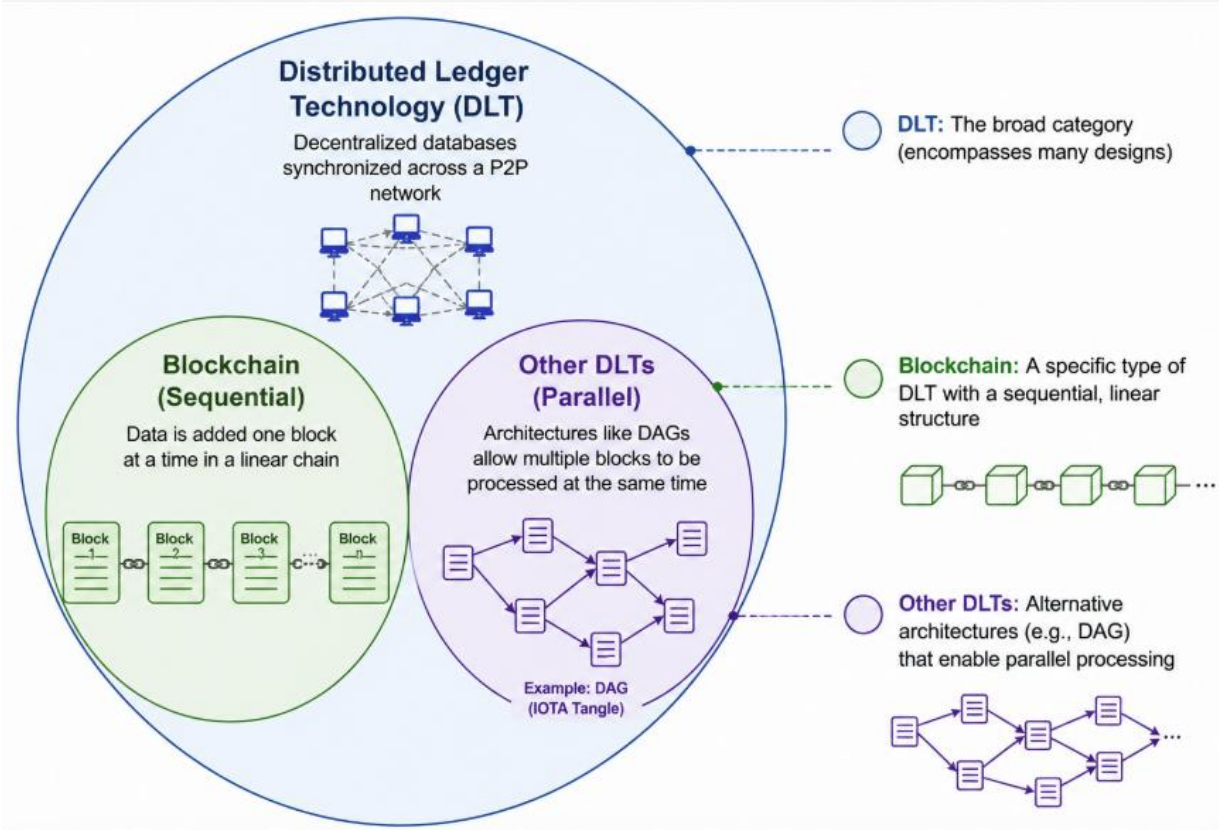
- ❖ Transactions are grouped into units called **blocks**
- ❖ Each block is linked to a previous block
- ❖ Blocks are ordered chronologically
- ❖ The first block in a blockchain is called the **genesis** block



1. What is a Blockchain?

Distributed Ledger Technology (DLT) vs. Blockchain:

- ❖ DLT: A broad category of decentralized databases synchronized across a P2P network
- ❖ Blockchain (Sequential): A specific type of DLT where data is added one block at a time in a linear chain



2. Characteristics of Blockchain

- ❖ Distributed nature
- ❖ Decentralized
- ❖ Immutable ledger
- ❖ Tamper-evident
- ❖ Pseudo-anonymity
- ❖ Provenance

2. Characteristics of Blockchain

Distributed nature

❖ Peer-to-peer (P2P) network

- Reliability
- Greater security
- No single point of failure
- Reduce cost
- Nodes act as server as well as client
- Distributed trust

❖ Distributed ledger

- Redundancy, shared and synchronized transactions/blocks

2. Characteristics of Blockchain

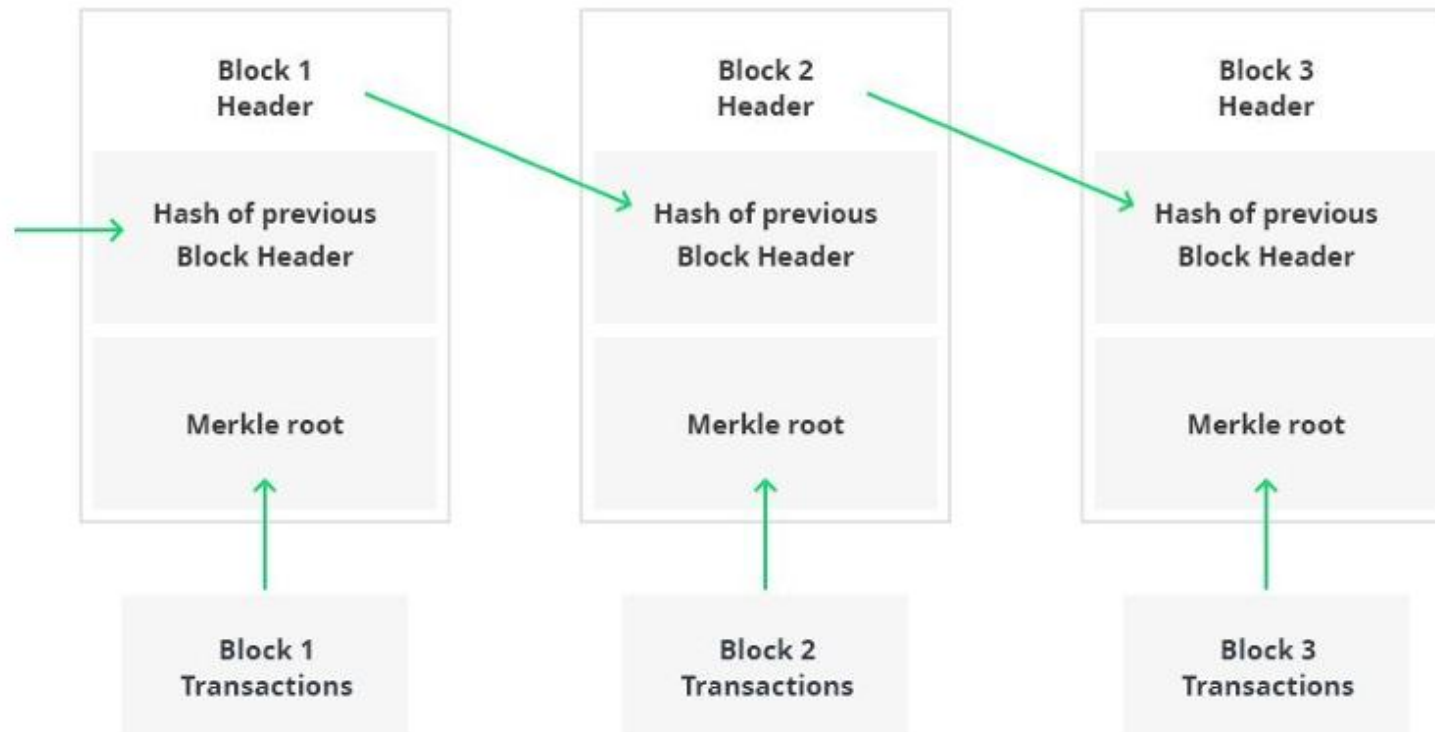
Decentralized

- ❖ Control is shared among many participants instead of a single authority.
- ❖ No trusted-third party to settle the transaction between two parties
- ❖ Transparent

2. Characteristics of Blockchain

Immutable ledger

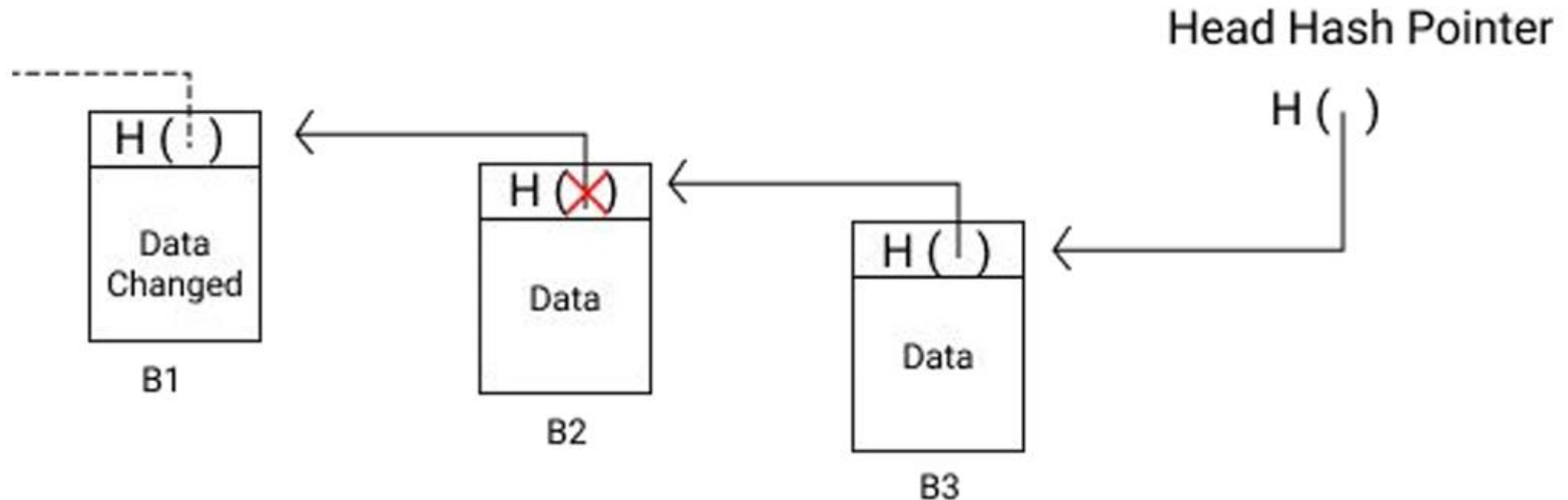
- ❖ Append only
- ❖ Data cannot be changed without network agreement



2. Characteristics of Blockchain

Tamper-evident

- ❖ Minor change in a transaction will invalidate the complete ledger



2. Characteristics of Blockchain

Pseudo-anonymity

- ❖ Blockchain is not anonymous
- ❖ Pseudo-anonymity
 - Disguised identity
 - Linked to public address
 - Transactions are public and identities may be inferred.
- ❖ No one will get to know the person actual name or other sensitive information
 - Increase confidentiality



2. Characteristics of Blockchain

- ❖ **Provenance:** An immutable historical record of who made changes and when
- ❖ Data provenance is a historical record
- ❖ Originates who made changes to it over time
- ❖ It validates if product/service is authentic
- ❖ Improve quality control and auditing

3. Cryptocurrencies

- ❖ Cryptocurrency is an application of blockchain, not blockchain itself.
- ❖ Cryptography-based digital money/currency
- ❖ Exchanges of assets (funds) without third-party or centralized control authority (e.g., banks or other financial institutions)

Q: Is blockchain the same as cryptocurrency?



3. Cryptocurrencies

Bitcoin

- ❖ Bitcoin is the first successful application of blockchain technology.
- ❖ Satoshi Nakamoto authored in 2008 and implemented in 2009
- ❖ First implementation of blockchain technology
- ❖ Currently, Bitcoin is the most traded cryptocurrency.

3. Cryptocurrencies

Ethereum

- ❖ Vitalik Buterin invented in 2015
- ❖ The emergence of Ethereum took the blockchain to next level, usually relate to a Blockchain 2.0
- ❖ Smart contracts in their platform
- ❖ Ethereum introduces the Ether cryptocurrency

3. Cryptocurrencies

Altcoins: All the cryptocurrencies alternative to Bitcoin are called Altcoins

❖ Forks of Bitcoin

- Bitcoin XT
- Bitcoin cash
- Bitcoin Gold, etc

❖ Small changes in Bitcoin and launch own cryptocurrency

- Litecoin

❖ Defining own rules and Blockchain network

- XRP (Ripple)

❖ Focusing on some particular problem solution

- Storj - Decentralized cloud storage
- Augur - Decentralized prediction market platform

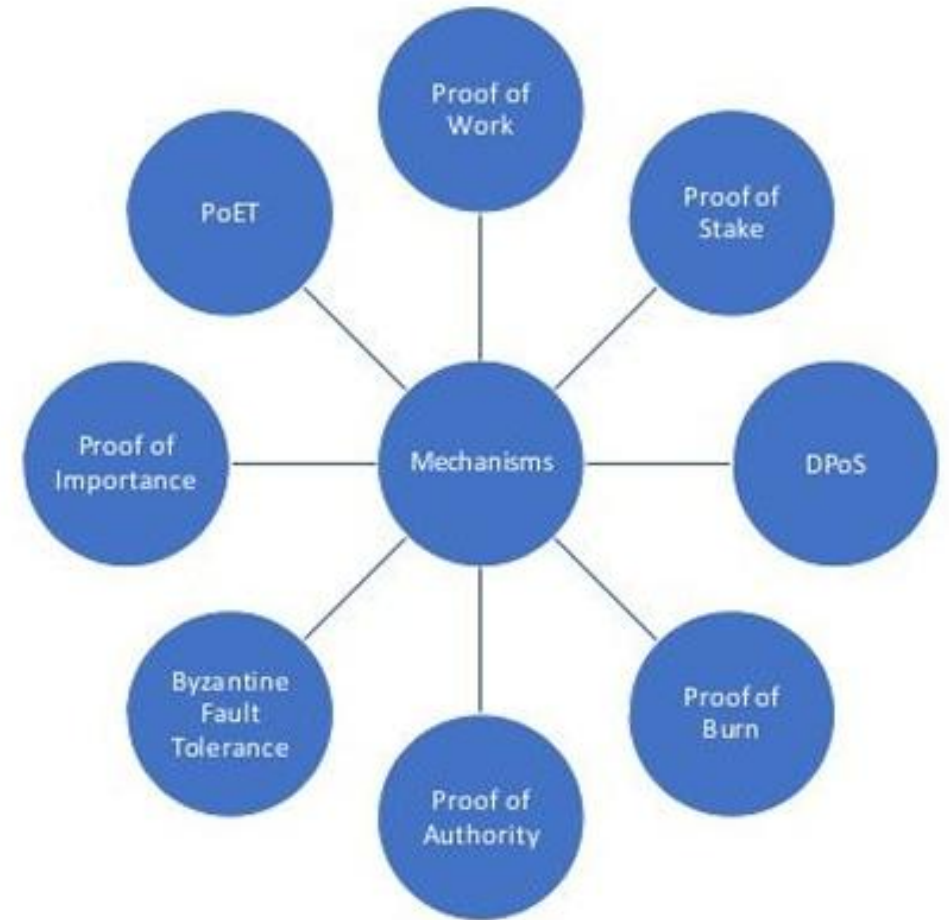


4. Consensus Mechanism



- ❖ Decision-making process
- ❖ Define the state of the ledger
- ❖ Dynamic way of reaching an agreement in a group

Q: Why do decentralized systems need consensus?



4. Consensus Mechanism

Proof of work (PoW)

- ❖ Mining-based consensus mechanism
- ❖ Bitcoin and Ethereum
- ❖ Energy-waste
- ❖ Try to solve computational puzzle



4. Consensus Mechanism

Proof of Stake (PoS)

- ❖ Hold/Lock crypto assets to participate in consensus process
 - Validators
 - Needs assets
 - Energy-efficient
 - Ethereum migrating to PoS



4. Consensus mechanism

Other consensus mechanisms

- ❖ Practical byzantine fault tolerance (PBFT)
- ❖ Delegated Proof of Stake (DPoS)
- ❖ Proof of Elapsed Time (PoET)
- ❖ Proof of Authority (PoA)
- ❖ Crash Fault Tolerance (CFT)



5. Mining



Transactions are verified and added to the blockchain ledger.

- ❖ Nodes (Miner nodes, Full nodes, Validators nodes)
- ❖ Crypto economics
- ❖ Crypto puzzle
- ❖ Mining difficulty level
- ❖ Nonce and role of nonce



5. Mining

Crypto mining equipment

- ❖ Mining hardware to solve complex crypto puzzle.



6. Crypto Wallets

- ❖ Manage cryptographic keys (especially private keys)
- ❖ Create transactions to send and receive cryptocurrency
- ❖ Track the cryptocurrency balance
- ❖ Wallet types (based on key storage):
 - Paper wallet
 - Hardware wallet
 - Software wallet

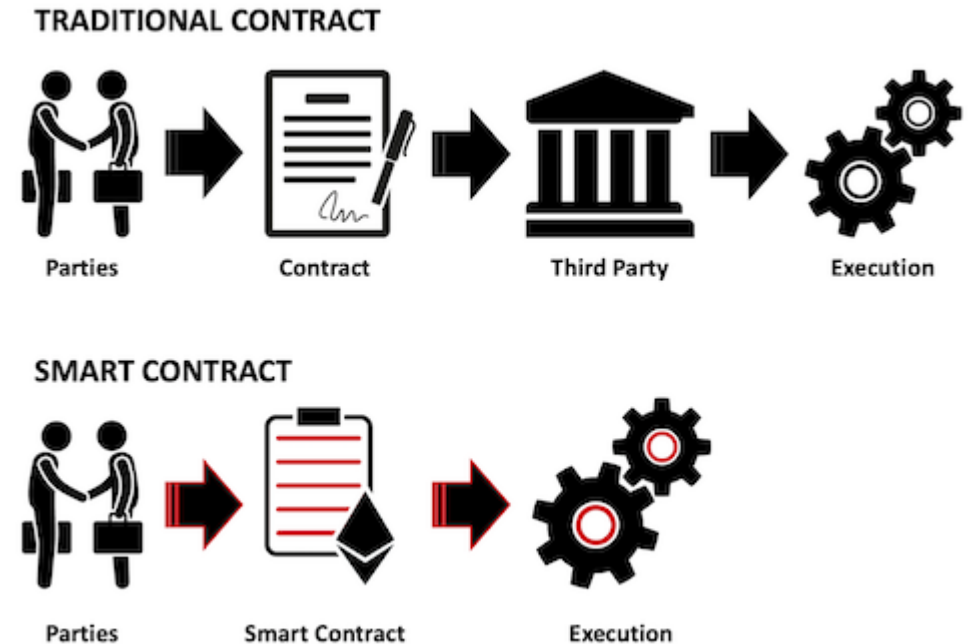


Q: Does a crypto wallet store your coins?

7. Smart Contracts

Smart contract is a digital contract

- ❖ **Piece of code** exists in a Blockchain network
- ❖ Execute when certain conditions meet
 - For example, self-execute when conditions/terms meet between buyer and seller
- ❖ Enables decentralized applications



8. Discussion



1. At a high level, how do consensus, mining, wallets, and smart contracts interact to process a transaction in a blockchain system?
2. Based on what you learned today, in what situations does using a blockchain make sense, and in what situations does it not?
3. If a Distributed Ledger (DLT) is shared among thousands of nodes, why is it usually slower than a single centralized SQL database?